PROBLEM SET 3, PART 2: TRANSERSALITY
DUE: OCT. 31

(1) [Smooth submanifolds as regular level sets]
Here is what we argued in class:

> 一方面，由子流形的定义可知，对于任意 $q \in X$，均存在邻域 V 以及由坐标卡映射所诱导的光滑映射 $g : V \to \mathbb{R}^l$（其中 $l = \dim N - \dim X$ 是 X 在 N 中的余维数）使得 $g^{-1}(0) = X \cap V$。注意由 g 的构造可知 dg 是满射，于是 0 是 g 的正则值，且它的核为
>
> $$\ker(dg_q) = T_q X.$$

(a) Summarize the above as a lemma.
(b) Show that the "global version" fails: Let K be the Klein bottle and let S be its central circle. Prove that there is no smooth function $f : K \to \mathbb{R}$ so that 0 is a regular value and $f^{-1}(0) = S$.
(c) (Not required) In general, suppose S is a smooth submanifold of M of codimension r, and suppose the normal bundle $N(S, M)$ is trivial in the sense that $N(S, M)$ is isomorphic(You may need to find the exact meaning of "isomorphic" between vector bundles) to $S \times \mathbb{R}^r$. Prove: There is a smooth map $f : M \to \mathbb{R}^r$ so that $0 \in \mathbb{R}^r$ is a regular value of f and $S = f^{-1}(0)$.
(d) (Not required) Conversely, suppose $f : M \to \mathbb{R}^r$ is a submersion, and $S = f^{-1}(0)$ is a submanifold of M. Show that $N(S, M)$ is trivial.
(e) Write down a theorem summarizing (c) and (d).

(2) [**Stability of various properties**]
We need the following definition:

> We say a property $\mathcal{P}$ concerning maps in $C^\infty(M, N)$ is a *stable property* if it is preserved under small deformation, namely, if $f \in C^\infty(M, N)$ satisfies $\mathcal{P}$ and F is a smooth homotopy with $F(x, 0) = f$, then there exists $\varepsilon > 0$ so that for each $0 < t < \varepsilon$, the map $f_t(\cdot) = F(\cdot, t)$ satisfies the property $\mathcal{P}$.

(a) Prove: If M is compact, then the following properties of maps in $C^\infty(M, N)$ are stable:
  (i) submersion,
  (ii) embedding.
(b) (Not required) Show that the conclusion above fails if M is non-compact by studying the following example.

> **Counterexample:** Let $h \in C^\infty(\mathbb{R})$ be a smooth bump function with
>
> $$h(x) = 1 \text{ for } |x| < 1, \qquad h(x) = 0 \text{ for } |x| > 2.$$
>
> Consider the map $F(x, t) = xh(tx)$ as a homotopy with $f_0(\cdot) = F(\cdot, 0)$.

(3) [**Stability of transversal intersection**]

(a) Show that if M is compact and X is a smooth submanifold of N, then the property "$f \in C^\infty(M,N)$ intersect X transversally" is a stable property.

(b) (Not required) Let $F : S \times M \to N$ be a smooth map. Suppose M is compact, and $X \subset N$ is a closed submanifold. Denote $f_s(\cdot) = F(s,\cdot)$. Prove: the set

$$\{s \in S \mid f_s \text{ intersect } X \text{ transversally}\}$$

is an open subset of S.

(4) [**Lefschetz maps**]

You will need the following conceptions.

Let $f : M \to M$ be a smooth map. A point $p \in M$ is a called a *fixed point* of f if $f(p) = p$. We say f is a *Lefschetz map* if for each fixed point p of f, 1 is not an eigenvalue of $df_p : T_pM \to T_pM$. The *local Lefschetz number* $L_p(f)$ of a Lefschetz map at a fixed point p is the sign of the determinant $\det(df_p - \mathrm{Id})$, i.e. $L_p(f) := 1$ if $\det(df_p - \mathrm{Id}) > 0$, and $L_p(f) := -1$ if $\det(df_p - \mathrm{Id}) < 0$.

Do the following questions:

(1) Let $r_\theta : S^2 \to S^2$ be the map "rotate S^2 by an angle θ", $(\theta \neq 2k\pi)$, defined by

$$r_\theta(x^1, x^2, x^3) = (x^1 \cos\theta - x^2 \sin\theta,\ x^1 \sin\theta + x^2 \cos\theta,\ x^3).$$

Prove: r_θ is a Lefschetz map.

(2) Let V be a vector space, and $L : V \to V$ a linear map. Let $\Delta = \{(v,v) : v \in V\}$ be the diagonal in $V \times V$, and $\Gamma_L = \{(v, Lv) : v \in V\}$ be the graph of L in $V \times V$. Prove: Γ_L intersects Δ transversally if and only if 1 is not an eigenvalue of L.

(3) Prove: If M is compact and $f : M \to M$ is a Lefschetz map, then f has only finitely many fixed points.

(4) The Lefschetz number of a Lefshetz map f is defined to be $L(f) = \sum_{f(p)=p} L_p(f)$, where the summation is over all fixed points p. Compute $L(r_\theta)$for r_θ in (1).

(5) [**Simply connectedness of $\mathbb{R}^n \setminus M$ ($\dim M \leq n-3$)**] (Not required)

Let M be a connected smooth manifold of dimension m. Prove: if $S \subset M$ is a smooth submanifold of dimension $k \leq m-3$, then for any $p \in M - S$, $\pi_1(M,p) \simeq \pi_1(M-S,p)$.